

When Do We Panic?

Cryptography in the shadow of the quantum computer — the threat, the mathematics, and why your browser is already hedging its bets.



WRITTEN & ILLUSTRATED BY

Ratty2Tails

On sound money, sound machines, and the mathematics that keeps a secret.

TL;DR

A large enough quantum computer would break the public-key cryptography that underpins nearly everything we do online — the RSA and elliptic-curve schemes behind secure browsing, signed certificates and key exchange. No such machine exists today. The largest error-corrected devices sit at roughly fifty qubits; breaking a real RSA key would take something like four thousand. The honest timeline, if it happens at all, is measured in decades.

So why prepare now? Because of *harvest now, decrypt later*: an adversary can record encrypted traffic today and wait for the machine to arrive. Anyone with secrets that must stay secret for thirty years should already be moving. Quietly, most of us already have — modern browsers run a *hybrid* key exchange that pairs a classical elliptic curve with a new lattice-based scheme, ML-KEM. Post-quantum cryptography does not mean we are post-quantum. It means we are insuring against a future in which we might be.

~50

ERROR-CORRECTED QUBITS
IN TODAY'S LARGEST MA-
CHINES

~4,000

QUBITS NEEDED FOR
SHOR'S
ALGORITHM TO BREAK RSA

2016

YEAR NIST OPENED ITS
POST-QUANTUM COMPETI-
TION

2^x

KEY EXCHANGES RUN AT
ONCE
IN MODERN HYBRID TLS

SCOPE

This is a field guide, not a specification. It explains why cryptographers are worried about quantum computers, which schemes are actually at risk, and what is being done about it — drawing on the state of play as understood in early 2026. It is deliberately light on equations and heavy on intuition. Where a claim rests on a machine that does not yet exist, this essay hedges accordingly. Nothing here is security advice for a specific system; it is a map of the terrain.

I The Insurance Argument

It is tempting to open a piece like this with the word *panic*. Resist it. The interesting thing about post-quantum cryptography is not that a machine has arrived to shatter our secrets — it is that no such machine has, and yet serious people are quietly rebuilding the foundations of digital security anyway. The correct posture is not alarm. It is something closer to the calm with which a sensible person buys house insurance.

The odds that your house burns down this year are small. You buy the policy regardless, because the cost of being wrong is catastrophic and irreversible. Cryptography now faces the same arithmetic. There is a small but non-negligible chance that, some decades hence, a quantum computer of sufficient size will exist. If it does, a great deal of the cryptography protecting the internet today would fall in an afternoon. The rational response is not to wait for the smoke — it is to fit the alarms while the house is quiet.

Post-quantum cryptography does not mean we are now post-quantum. It means we are preparing for a world in which we might be.

THE CENTRAL IDEA, IN ONE LINE

Two things follow from taking the threat seriously without succumbing to it. First, we must be precise about *what* is actually at risk — because, as we shall see, the quantum threat is oddly selective. Second, we must accept that the replacements are themselves young and unproven, which is why the transition is being done with belt and braces rather than a single leap of faith.

What the quantum computer actually threatens

The danger is selective. Public-key schemes fall; symmetric ciphers and hashes merely need larger keys.

BROKEN BY SHOR'S ALGORITHM		MERELY WEAKENED BY GROVER'S	
RSA Integer factorisation	BROKEN	AES-128 Search space $2^{128} \rightarrow 2^{64}$	USE 256-BIT
Elliptic-Curve Crypto (ECC) Elliptic-curve discrete log	BROKEN	AES-256 Search space $2^{256} \rightarrow 2^{128}$	SAFE
Diffie-Hellman key exchange Discrete logarithm problem	BROKEN	SHA-256 & hash functions Only square-root speed-up	SAFE
<i>These schemes rest on problems a quantum computer solves in polynomial time. They must be replaced entirely.</i>		<i>Grover only square-roots the search. Double the key length and the problem retreats safely out of reach again.</i>	

Figure 1. Not all cryptography is equally exposed. Public-key schemes — the ones that establish trust and exchange keys — are the casualties. Symmetric ciphers and hash functions survive by simply moving to larger keys.

II Two Algorithms, Two Threats

A quantum computer is not merely a fast classical computer. That distinction cannot be repeated often enough. It runs a wholly different repertoire of algorithms, most of which are useless for most problems. The quantum threat to cryptography comes down to just two of them — and they are not equally dangerous.

GROVER'S ALGORITHM — THE MANAGEABLE ONE

Grover's algorithm attacks unstructured search. If breaking a cipher means guessing a key by brute force, Grover takes the square root of the work. Against a 128-bit AES key, a classical attacker faces 2^{128} guesses; Grover brings that down to 2^{64} , which is uncomfortably close to feasible. But the fix is almost insultingly simple: use a longer key. AES-256 sits at 2^{256} , and Grover reduces it only to 2^{128} — a number that remains beyond any computer, quantum or classical, for the foreseeable future. Symmetric cryptography shrugs and moves on.

SHOR'S ALGORITHM — THE REAL PROBLEM

Shor's algorithm is the one that keeps cryptographers awake. It factors large integers in polynomial time and solves discrete logarithms too. Take a large number n that is secretly the product of two primes, p and q . On a classical machine, recovering p and q from a 2,000- or 4,000-bit number is so slow as to be hopeless — that slowness is the security of RSA. Shor makes it fast. And with fast factoring and fast discrete logs, an adversary could forge certificates, hijack sessions, and decrypt anything ever secured with these schemes. This is not a weakening. It is a collapse.

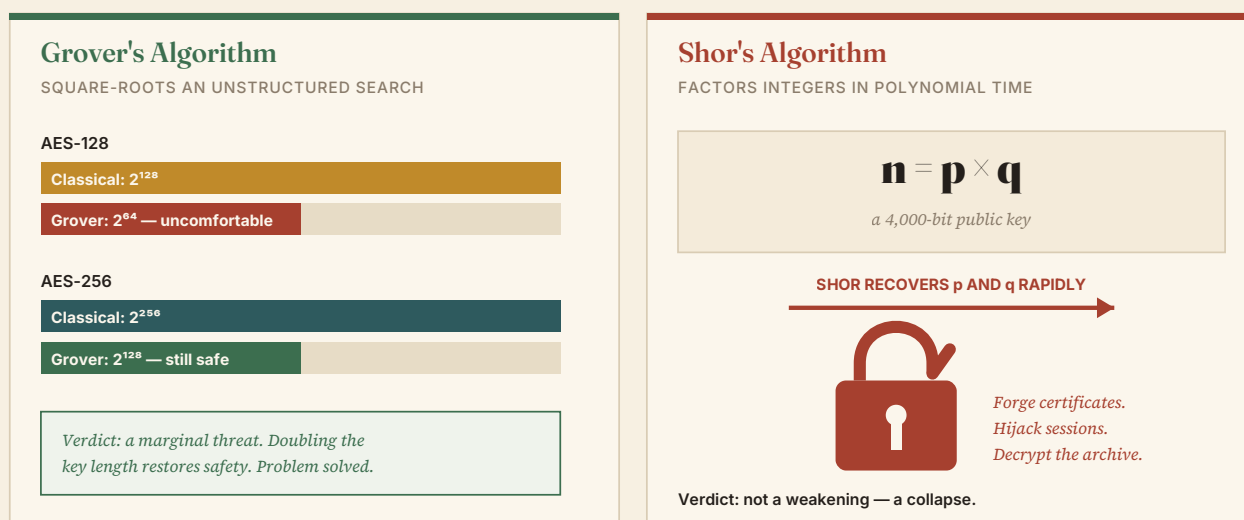


Figure 2. The two quantum algorithms that matter. Grover nibbles at symmetric ciphers and is defeated by longer keys. Shor dismantles the public-key schemes outright — which is why the whole transition effort is aimed squarely at replacing them.

III Harvest Now, Decrypt Later

If the machine is decades away, why move today? The answer is a strategy with an unsettling name: *harvest now, decrypt later*. An adversary need not wait for the quantum computer to *use* your data — only to *read* it. So they capture your encrypted traffic now, write it to a very large disk, and file it away. When the machine finally arrives, they decrypt the lot at their leisure. The secret you send today is only as safe as the cryptography will be for as long as that secret matters.

Which raises a fair question: what use is twenty-year-old information? For most of us, very little. By the time any quantum computer could crack it, the credit-card number I typed into a shop has expired, and my passport has been renewed. The threat is not democratic. It falls hardest on

those whose secrets keep their value for decades — governments, militaries, intelligence services, and anyone whose confidential records must remain confidential well into the 2050s. As it happens, those are precisely the organisations already drawing up transition timelines.

The long patience of an adversary



Figure 3. Why the timeline is not an excuse to delay. Data recorded today can be broken years later. Secrets with a long shelf life are already exposed — the machine simply hasn't collected on the debt yet.

IV The State of the Machines

How close, then, is the machine? Not close. The largest quantum computers built to date operate at roughly fifty error-corrected qubits — qubits with enough resistance to the noise of the system to be genuinely useful. Running Shor's algorithm against a real RSA key would demand something on the order of four thousand. That is not a gap to be closed with a bit more funding and a good quarter; it is a difference in kind.

Estimates for when — or whether — we cross that gap run to decades. It is entirely reasonable to remain sceptical that we ever will. And yet scepticism about the machine is no argument against preparation, for the same reason that doubting a fire is no argument against insurance. We do not build post-quantum schemes because the threat is certain. We build them because it is *possible*, and because the replacement, if we choose well, will simply shrug at the machine when — or if — it arrives.

The gap between now and the threat

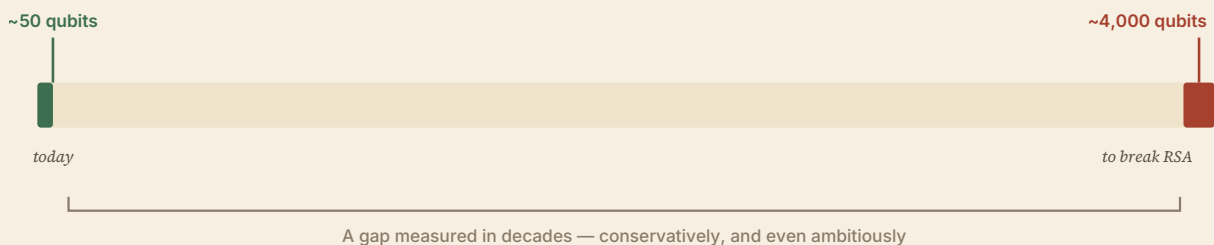


Figure 4. The distance still to travel. Today's best machines fall roughly two orders of magnitude short of the scale that would threaten RSA. Whether the gap ever closes is genuinely unknown — which is exactly the uncertainty post-quantum cryptography is designed to survive.

A NOTE ON NAMES

RSA & ECC — the public-key workhorses, securing certificates and key exchange. Both broken by Shor. **Kyber / ML-KEM** — the same lattice-based scheme under two names: "Kyber" was the competition entry; "ML-KEM" is its standardised form. **KEM** — a Key Encapsulation Mechanism, the modern way two parties agree a shared secret. **LWE** — Learning With Errors, the hard lattice problem underneath it all.

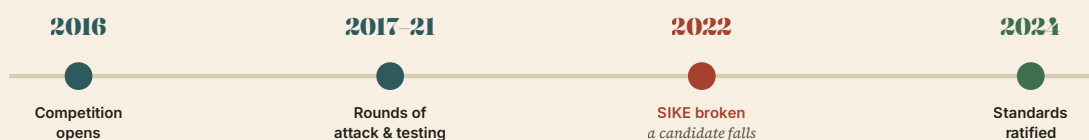
v The Search for Quantum-Proof Mathematics

Since the danger is specific — a handful of quantum algorithms against a handful of number-theoretic problems — the defence is specific too. We do not need cryptography that beats a magically fast computer in general; no such thing is possible. We need schemes resting on problems for which no efficient algorithm is known, classical *or* quantum. Finding and vetting those has been the work of the last decade.

The effort has been driven, in large part, by NIST — the United States' National Institute of Standards and Technology — which opened a public competition in 2016. Candidate schemes were submitted, tested, and attacked over successive rounds, and a set of winners has since been ratified as standards. It has not been smooth sailing. One key-exchange candidate, an elegant scheme called SIKE, was demolished in 2022 by a classical attack that exploited a deep structural flaw. It could easily have been one of the chosen. That it wasn't is a mark of the process working — but also a reminder that these schemes are young, and that confidence must be earned slowly.

Some of these algorithms are very new, and largely untested. We cannot be sure yet — which is exactly why we roll them out with such care.

THE CAUTIONARY PRINCIPLE

A decade of vetting

The winners — ML-KEM for key exchange, and companion signature schemes — emerged battered but standing.

Figure 5. Standardisation by ordeal. A scheme earns trust not by being clever but by surviving years of concerted attack. SIKE's collapse in 2022, mid-competition, is the clearest argument for patience.

vi Lattices and Hashes

Two families of hard problem now dominate the post-quantum landscape, and it is worth understanding *why* they are trusted — because the reason is more honest than it is reassuring.

LATTICES — THE WORKHORSE

Picture a regular grid of points marching off in every direction: a lattice. Now imagine I hand you a point floating somewhere off the grid and ask which lattice point lies nearest. In two dimensions this is trivial — you simply look. But the lattices used in cryptography have not two dimen-

sions but a thousand, and you cannot see them. The problem of finding the nearest point, or the shortest vector, or of solving a close cousin called *Learning With Errors*, becomes ferociously hard. Crucially, it is not merely hard for classical computers; we know of no quantum algorithm that cracks it either. Lattices are not chosen because they are better than elliptic curves. They are chosen because, so far as anyone can tell, they do not yield to a quantum trick.

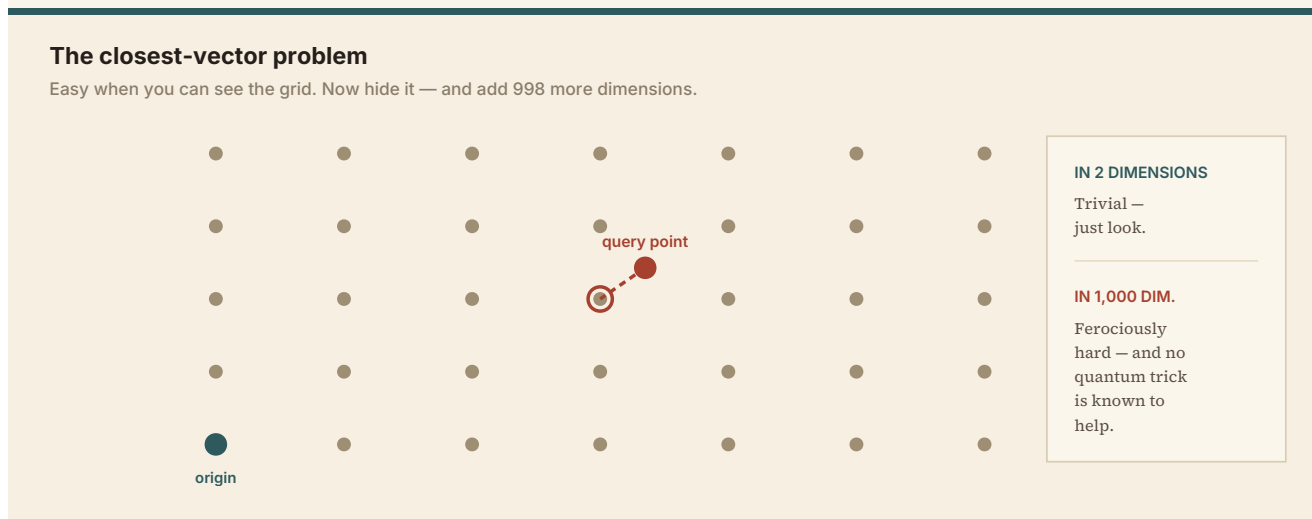


Figure 6. The intuition behind lattice cryptography. What is obvious on a flat grid becomes intractable in a thousand dimensions you cannot picture. That intractability — for classical and quantum machines alike — is the security.

HASHES — THE OLD FAITHFUL

The second family leans on hash functions, those one-way workhorses we already trust and understand. From the one-way property alone, one can build signature schemes: publish a set of hashes, and later reveal the original inputs as proof that the signature is yours. It is a gross simplification, but the shape is right. These hash-based schemes inherit two comforting facts. We already know hashes resist classical attack, and — thanks to Grover being merely a square-root speed-up — a 256-bit hash sits comfortably beyond quantum reach too. Anything built on hashes is likely to still be standing in thirty years.

A note of intellectual honesty is due here. These are not magic. They are simply algorithms for which no efficient attack, quantum or classical, is currently known. We do not have a proof that lattices are safe — only an absence of any known break. It is conceivable that someone will one day find the classical or quantum algorithm that undoes them. That lingering doubt is not a flaw in the plan. It is the reason for the plan's final, clever move.

VII Belt and Braces: The Hybrid Handshake

Because we do not fully trust the new schemes, we do not rely on them alone. Modern secure browsing performs *two* key exchanges at once. One uses a battle-tested classical algorithm — an elliptic curve such as X25519, the workhorse of the last decade. The other uses a post-quantum scheme — ML-KEM, the standardised form of Kyber, built on Learning With Errors. The two shared secrets are concatenated and fed together into a key-derivation function, and the session key is drawn from the combination.

The beauty of this arrangement is that it fails safe. To break the session, an attacker must break *both* exchanges. A future quantum computer might defeat the elliptic curve — but not the lattice.

A surprise mathematical break might one day undo the lattice — but the elliptic curve would still hold, at least against a classical adversary. Neither leg has to be perfect. They only have to fail in different ways.

The hybrid handshake, as your browser does it now

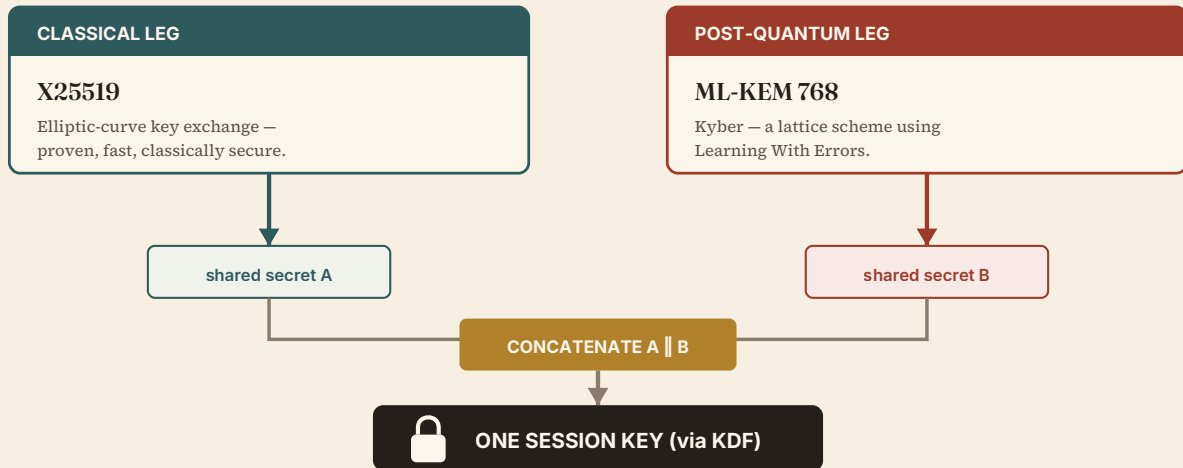


Figure 7. Belt and braces. Two independent key exchanges — one classical, one post-quantum — are combined into a single session key. Open a modern browser's security panel and this is very likely what you'll find already running. **To break it, an adversary must break both.**

And here is the quiet punchline. If you have visited Google, or most major sites, in a recent browser, you have almost certainly done this already — without noticing, without opting in, without a single headline. The people who insist that "quantum computers don't exist, so this is a waste of time" are half right. At this scale, they don't. But the algorithms that would still stand up when they do are no longer theoretical. They are running, today, on the machine in front of you.

Conclusion — Preparing for a world we might never inhabit

Post-quantum cryptography is a study in acting sensibly under deep uncertainty. The machine that would justify all this effort may be thirty years away; it may be a hundred; it may never arrive at all. A reasonable person can doubt it and still support the work — just as a reasonable person insures a house that will probably never burn.

What makes the field genuinely interesting is not the hardware. It is the way a conceivable-but-unlikely threat reshapes the decisions we make today. Harvest now, decrypt later turns tomorrow's machine into today's problem for anyone with lasting secrets. The response has been careful rather than panicked: a decade of public competition, a candidate scheme broken along the way as a warning, two families of hard mathematics we trust precisely because nobody has yet broken them, and a hybrid handshake that refuses to bet everything on any single idea.

We are not post-quantum. The point was never to declare victory over a machine that does not exist. The point was to build cryptography that will simply shrug — and go on keeping the secret — whether the machine arrives in twenty years, or never comes at all. The alarms are fitted. The house is still quiet. That is exactly when the sensible work gets done.

The Ratty2Tails Papers · The Machine Series, No. I. Written and illustrated by Ratty2Tails. Set in Fraunces, Source Serif 4 and Inter; diagrams hand-drawn as flat-fill SVG; typeset with WeasyPrint. This essay is an explanatory field guide drawn from a public discussion of post-quantum cryptography, reworked and expanded in the author's own words. Figures for the state of quantum hardware and standardisation reflect the picture as understood in early 2026 and are given with due editorial hedging. Not security advice for any specific system. © Ratty2Tails.